

Field Intelligence Report

Parcel 54000 · Generated 27 July 2026 · LPIS 2024



Selected parcel among surrounding mapped fields (illustrative)

Crop Detection

SATELLITE ESTIMATE

■ **Maize**

Confidence: **100%** ■ High

LPIS DECLARATION (FARMER)

■ **Barley - Spring**

Status: ■ **Satellite estimate differs**

This difference is presented as evidence for review, not as a determination that the declaration is incorrect — see Methodology for how differences are calculated.

1.6 ha

FIELD AREA

Silking

GROWTH STAGE (4/7)

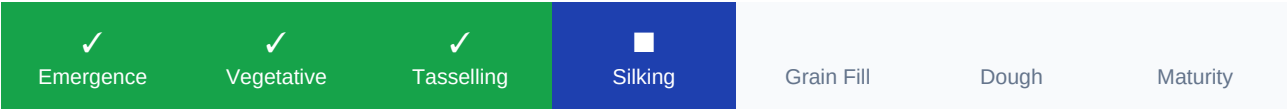
0.80

NDVI (VEGETATION)

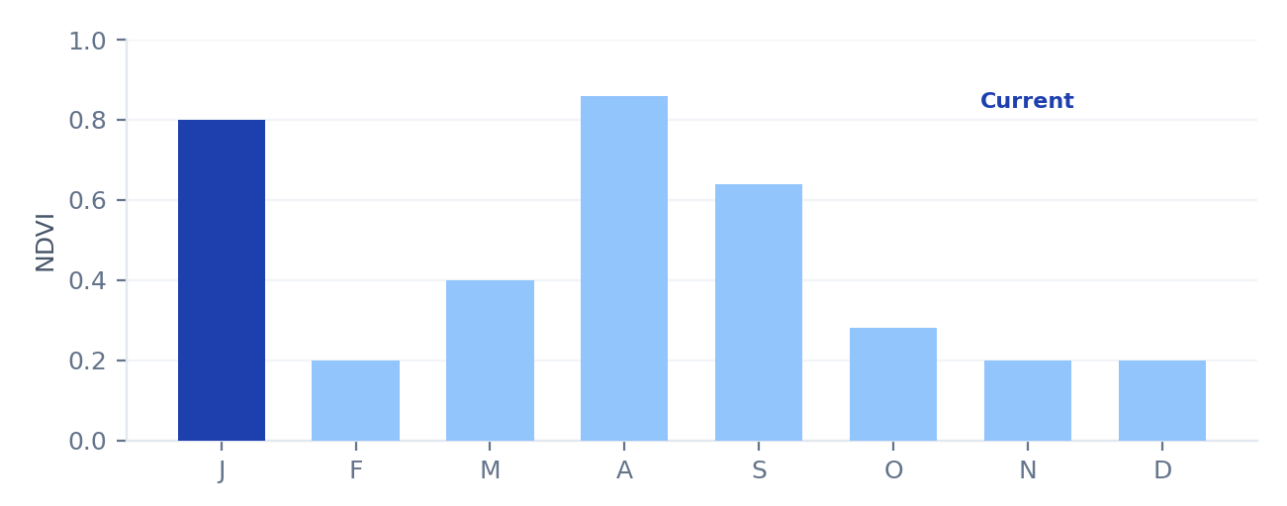
5–15 Oct

HARVEST WINDOW

Growth Timeline



NDVI Seasonal Profile



Satellite Observations

Metric	Value	Note
NDVI (Vegetation)	0.80	Dense canopy
Soil Moisture (estimate)	Medium (29%)	Live, per-location — Open-Meteo
Cloud cover (7-day)	~45%	Affects imagery freshness

7-Day Weather Forecast

6 mm EXPECTED RAIN	9–22°C TEMPERATURE RANGE	20 km/h WIND
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AI Insights

■ **High confidence difference from LPIS**

Satellite strongly suggests Maize but the LPIS declaration is Barley - Spring. This parcel may warrant ground verification.

■ **Healthy crop**

Vegetation indices indicate good canopy development. No signs of stress detected from satellite data.

■ **Next satellite pass**

HLS revisit in approximately 4 days. Updated NDVI and vegetation maps will be available.

Parcel Information

Area	1.6 ha (~4 acres)	Parcel ID	54000
Centroid Latitude	53.1602°N	Centroid Longitude	7.0502°W
Source	LPIS 2024	Data vintage	2022–2025 (4 seasons)

Methodology & Disclaimer

How this report was produced

This crop prediction is generated using a CatBoost machine learning model trained on 61,566 Irish tillage parcels from the LPIS 2024 dataset, using HLS (Harmonized Landsat Sentinel-2) satellite imagery — NDVI, EVI, NDWI, NDRE and NDII vegetation indices — across four growing seasons (2022–2025). The model achieves 77.05% balanced accuracy nationally on held-out validation data. Confidence scores are calibrated using isotonic regression so that the displayed percentage reflects genuine prediction reliability. Soil moisture is fetched live per-location from Open-Meteo at the time each parcel is viewed.

Reading confidence scores

■ High (80–100%)	Prediction is usually reliable.
■ Medium (50–79%)	Similar crops may also fit the satellite signature.
■ Low (<50%)	Use as a prompt for further review, not as a conclusion.

Known limitations

- Spring cereals (Barley, Oats, Wheat) are more easily confused with each other than with Maize or winter crops.
- Satellite observations are affected by cloud cover; confidence may decrease when limited clear imagery is available.
- Growth stage and harvest window are model estimates, not field-verified dates.
- The current model covers 8 tillage crop types only; grassland and other land uses are not classified.
- Soil moisture is a live estimate from weather-model data, not a direct field sensor reading.

Disclaimer: Cube Earth provides decision-support information and should not be used as the sole basis for regulatory or contractual decisions without additional verification. A difference between satellite prediction and LPIS declaration does not, by itself, establish that either is incorrect.

